

Encouraging Water Conservation in Schools: Testing the Impact of Awareness Campaigns on Grade 7 Students' Water Usage Habits

By

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An action research report submitted in partial fulfillment

of the requirements for the degree of

B.Ed. (Hons) Elementary

GOVERNMENT COLLEGE OF EDUCATION F.B. AREA,

KARACHI

(2021, 2025)

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Supervised

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"Read, and your Lord is the most Generous-Who taught by the pen-taught man that which he
knew not."

(Surah Al-Alaq 96:3-5)

"The superiority of the scholar over the worshipper is like that of the moon over all other
heavenly bodies."

(Sunan Abu Dawood, Hadith 3641)

"Whoever treads a path in search of knowledge, Allah will make easy for him a path to
Paradise."

(Sahih Muslim, Hadith 2699)

Certificate of Approval

I, Maleeha Khan D/o Mustajeeb Ahmed Khan student of B.Ed. (Hons) Elementary has successfully completed the Action Research entitled "Encouraging Water Conservation in Schools: Testing the Impact of Awareness Campaigns on Grade 7 Students Water Usage Habits". The study fulfills the scholarly standard set by the research project course guide/University of Karachi.

Signature of the Candidate:

Name of Candidate: Maleeha Khan

Signature of the Supervisor:

Name of the Supervisor: Shagufta Khokhar

Declaration

I, Maleeha Khan D/O Mustajeeb Ahmed Khan, student of B.Ed. (Hons) Elementary hereby declare that no part of this thesis titled " Encouraging Water Conservation in Schools: Testing the Impact of Awareness Campaigns on Grade 7 Students' Water Usage Habits" presented by me has been plagiarized from anywhere. Proper references are cited wherever needed.

I understand that the University of Karachi reserves the right to cancel the degree if any of the above declaration is proved false before or even after award of degree.

Signature of the Candidate: __

Name of Candidate: Maleeha Khan

Seat Number: A1-21136058

Dedication

I dedicate this action research to my beloved Mother Afsheen Anwer whose unwavering support has been the foundation of my journey. My Mother's dreams for my success continue to inspire me every day. Her belief in my potential and encouragement guide me through every challenge. Her immense support and love have been my greatest strength throughout my academic journey. My mother's faith in me fuels my passion for learning and growth and for that, I am forever grateful. This work is a reflection of her love and support and I dedicate it to her with all my love.

Acknowledgement

Firstly, I want to express my gratitude to Allah (S.W.T) for giving me the strength to complete my research project. I would like to express my sincere gratitude to my supervisor Prof. Shagufta Khokhar, for her immense support and invaluable guidance throughout this research project.

This research project would not have been possible without her mentorship. I am truly grateful to have had the chance to learn from her.

I also want to thank my dear fellow Fatima Siddiqui and my all the subject teachers who have supported me throughout my B.Ed. journey. Their guidance and motivation played a significant role in shaping my understanding and passion for learning.

Abstract

Pakistan is experiencing severe water scarcity, yet environmental education in government schools remains largely textbook-driven, examination-oriented, and disconnected from students' lived ecological realities. Soomro et al. (2025) and Bibi et al. (2025) highlighted that school teach environmental topic but student rarely do inquiry based activities, they just learn the concept but do not develop the skills to act on it. Similarly, another studies also shared that experiential learning and specifically campaign-based action oriented environmental education, improves pro environmental behaviors, builds civic responsibilities and long-term ecological stewardship in young children's (Ardoin et al., 2020; UNESCO, 2021; Tilbury, 2018). However, in Pakistan, low-resource public schools have limited use of these active learning approaches, and this area is not well studied. Therefore, this qualitative action research addressed this gap by observing the effectiveness of an awareness-campaign methodology for building water-saving habits among 20 students of seventh-grade girls at a government girls secondary school in Karachi. Grounded by Creswell's cyclical Action Research model, three experiential learning cycles were implemented on six lessons that incorporate Science and Social Studies concepts, students led campaign activities, engaged in positive discussions, created awareness campaign promotion material, and motivational messages to their families and school community. Data from semi-structured teacher interviews, focus groups with students, reflective journals, and lessons were thematically analyzed.

Baseline findings revealed the incomplete understanding, limited environmental authority, and the control of rote teaching strategies. Post intervention proved the effective impact on students by demonstrating enhance critical thinking, understanding, confident public speaking and unleash their creativity as well transform young students in an agents of positive change. Teachers stated a methodological shift towards learner-centered strategies.

This Action Research participates to Pakistani and global scholarships by displaying the impact of campaign-based, experiential and critical strategies can shift environmental learning exceed awareness towards sustained behavioral and promote positive change in community.

Key Words

Severe water scarcity, Experiential learning, Methodological shift, Environmental education, Awareness-campaign methodology, Promote positive change, and Textbook-driven

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Chapter 01 Introduction

1.1 Introduction

Water is an essential natural resource, vital for sustaining life, supporting ecosystems, and facilitating economic development. However, Pakistan faces a severe water crisis, with per capita water availability declining from 5,260 cubic meters in 1951 to less than 1,000 cubic meters in recent years, categorizing the country as water-scarce (Pakistan Council of Research in Water Resources, 2020). Factors contributing to this crisis include rapid population growth, urbanization, inefficient water use, and climate change.

Educational institutions play a pivotal role in shaping the attitudes and behaviors of young individuals. Integrating water conservation education into the school curriculum can instill a sense of responsibility and promote sustainable practices among students. By combining scientific knowledge with social awareness, students can comprehend the multifaceted nature of water issues and become proactive in addressing them.

This study focuses on implementing awareness campaigns within Grade 7 classrooms, integrating Science and Social Studies concepts to enhance students' understanding and encourage water-saving behaviors.

1.2 Statement of the Problem

Despite the critical nature of water scarcity in Pakistan, there is a noticeable gap in environmental education, particularly concerning water conservation, within the primary school curriculum. Many students lack awareness of the severity of water issues and the importance of conservation practices. Traditional teaching methods often fail to engage students in meaningful discussions about environmental challenges, leading to a disconnect between knowledge and action.

The absence of structured awareness campaigns and interactive learning experiences contributes to students' indifferent attitudes toward water usage. This research seeks to address this gap by implementing targeted awareness campaigns and evaluating their impact on students' knowledge, attitudes, and behaviors related to water conservation.

1.3. Significance of the Study

This study holds significant importance in addressing both educational and environmental challenges faced by Pakistan. Firstly, from an educational perspective, it contributes to enhancing the primary school curriculum by integrating environmental education, particularly focusing on water conservation — an area often overlooked in Pakistani textbooks and classroom instruction (RafiUllah et al., 2018). By combining concepts from Science and Social Studies, this research fosters interdisciplinary learning and critical thinking skills, encouraging students to analyze real-world problems from multiple perspectives.

Secondly, the study aims to bring about behavioral change among Grade 7 students by instilling sustainable water usage habits. Developing these habits at an early age is crucial, as it lays the foundation for long-term environmental stewardship and responsible citizenship (Fayyaz et al., 2023). Through awareness campaigns and interactive activities, students are expected to become more conscious of their daily water consumption and adopt practices that contribute to water conservation in their homes and schools.

Thirdly, this research aligns closely with national environmental priorities and policy frameworks. It supports initiatives like the Clean Green Pakistan Movement, launched by the Government of Pakistan to promote eco-friendly behaviors and sustainable living (Ministry of Climate Change, 2019). Additionally, it complements Sustainable Development Goal 6 (Clean Water and Sanitation) by raising awareness about water scarcity issues and encouraging conservation efforts at the grassroots level (UNDP Pakistan, 2022).

Moreover, the study promotes community engagement by motivating students to act as ambassadors of water conservation within their neighborhoods and families. This ripple effect has the potential to create a broader societal impact, as students disseminate water-saving practices learned through school campaigns to the wider community, an approach previously proven effective in local environmental education programs (WWF-Pakistan, 2020).

Finally, the research makes a valuable contribution to the academic field by providing empirical evidence on the effectiveness of awareness campaigns in influencing students' environmental attitudes and behaviors. It fills a notable gap in local scholarship, where limited studies have explored the integration of environmental education with behavioral outcomes in the context of Pakistani primary schools (Bukhari & Kamal, 2017). The findings of this study could inform future curriculum development, teacher training programs, and school-based environmental initiatives across the country.

1.4 Research Question

What is the current level of water conservation awareness among Grade 7 students?

How effective are awareness campaigns in improving students' knowledge and habits related to water conservation?

What changes in water usage behavior can be observed after implementing awareness campaigns in schools?

1.5 Research Objectives

To assess the initial level of water conservation awareness among Grade 7 students.

To develop and implement a water conservation awareness campaign combining Science and Social Studies concepts.

To evaluate the impact of the campaign on students' knowledge, attitudes, and daily water usage habits.

To promote sustainable water management practices within the school community.

1.5 Definitions of key terms

Water Conservation The practice of using water efficiently to reduce unnecessary water usage.

Awareness Campaign

A series of activities designed to inform and educate individuals about a particular issue, aiming to influence attitudes and behaviors.

Sustainable Development Goals (SDGs)

A collection of 17 global goals set by the United Nations to address pressing environmental, political, and economic challenges.

Action Research

A participatory study conducted by individuals within their own work environment to improve practices and outcomes.

Chapter 2 Literature Review

2.1 Water Scarcity in Pakistan

Pakistan is currently confronting one of the most critical water shortages in its history.

According to the **World Bank (2019)**, the nation is projected to face absolute water scarcity by 2025, making Pakistan one of the most water-stressed countries globally. Various factors contribute to this crisis, including rapid urbanization, population growth, inefficient agricultural practices, climate-change-induced fluctuations in rainfall, and excessive groundwater extraction. The **Pakistan Council of Research in Water Resources (PCRWR, 2020)** reports that major cities such as Karachi and Lahore are already experiencing dangerous groundwater depletion. Similarly, the **Ministry of Climate Change (2021)** warns that nearly 40% of Pakistan's water supply is lost due to leakage, wastage, and mismanagement. International bodies like **UNESCO (2021)** describe Pakistan's water situation as alarming, stressing that without immediate educational and behavioral interventions, the crisis will escalate further.

2.2 Environmental Education in Pakistani Schools

Environmental education has gradually gained importance in Pakistan, particularly in response to environmental degradation and climate challenges. The **Clean Green School Programme**, initiated by **WaterAid Pakistan** in collaboration with the Federal Directorate of Education, emphasizes action-based learning and climate literacy. Likewise, the **WWF-Pakistan Green School Programme (2020)** engages students in hands-on conservation activities, environmental clubs, and awareness sessions to build environmental responsibility from an early age.

However, despite these promising efforts, several challenges persist. **RafiUllah et al. (2018)** identified major inconsistencies in environmental content across Pakistani textbooks, especially at elementary and secondary levels. Moreover, **UNICEF (2020)** reports that many schools rely heavily on lecture-based teaching, lacking opportunities for student-led inquiry and practical engagement. Similarly, **UNESCO (2017)** notes that environmental education in Pakistan usually emphasizes theoretical knowledge but rarely integrates cognitive, affective, and psychomotor domains together. Consequently, students often struggle to translate knowledge into sustainable behavior, revealing a significant gap in current educational practices.

2.3 Awareness Campaigns and Behavioral Change

Awareness campaigns are widely recognized as effective tools for shaping environmental behaviors. Research conducted in Karachi by **Fayyaz et al. (2023)** found that eco-friendly school initiatives significantly improved students' conservation habits and environmental awareness. Similarly, studies by the **International Water Management Institute (IWMI, 2020)** demonstrated that school-based water-saving campaigns contributed to noticeable improvements in household water-use practices, influenced by children's advocacy.

Globally, studies reinforce the importance of awareness campaigns for environmental learning. **UNESCO (2019)** explains that long-term behavior change is more likely when students are actively involved in hands-on experiences rather than passive listening.

UNICEF (2020) reports significant improvements in water-saving habits among Asian schoolchildren who participated in WASH campaigns that included demonstrations, games, and parental involvement. These findings highlight the importance of interactive, student-centered awareness campaigns in fostering meaningful behavioral change.

2.4 Integration of Science and Social Studies for Environmental Learning

Integrating knowledge from different subjects enhances students' understanding of environmental issues. Science provides foundational knowledge of the water cycle, pollution, evaporation, hydrology, and climate change, while Social Studies connects these scientific phenomena to human activities, resource management, society, and culture. This interdisciplinary approach is strongly recommended by **UNESCO's Education for Sustainable Development (ESD) framework (2018)**, which stresses that environmental problems cannot be effectively understood through isolated subjects.

In the Pakistani context, however, research by the **Higher Education Commission (2020)** indicates that environmental concepts are often taught in isolation, limiting students' ability to connect scientific knowledge with real-world social and cultural contexts. This gap calls for innovative teaching approaches that blend Science and Social Studies content, encouraging students to apply knowledge to practical and societal issues.

2.5 Importance of Environmental Education for Children

International research shows that children can act as powerful “environmental ambassadors.” **Ballantyne et al. (2006)** found that when children learn environmental values at school, they often transfer these habits to their families and communities. **UNESCO (2020)** also argues that developing environmental responsibility at a young age fosters empathy, critical thinking, and sustained environmentally friendly behavior. Given Pakistan's water crisis, empowering children through school-based education is essential for future sustainability.

2.6 Research Gaps in Existing Literature

While numerous initiatives and studies highlight the importance of environmental education, several gaps remain—particularly within the Pakistani school context.

First, existing school programs tend to be teacher-centered, offering limited opportunities for students to take leadership roles. In contrast, the present action research prioritizes **student-led involvement**, allowing students to suggest ideas, lead discussions, create awareness materials, and engage in decision-making.

Second, previous studies largely focus on general awareness but seldom develop **critical thinking**. This research addresses that gap by employing **Paulo Freire's critical pedagogy**, encouraging inquiry-based dialogue, problem-solving, and reflective thinking among students.

Third, current environmental programs often focus only on knowledge (cognitive domain), ignoring emotional (affective) and practical (psychomotor) development. The present study explicitly targets **all three domains**, enabling students to understand water issues, develop empathy through Islamic perspectives, and apply skills through hands-on conservation activities.

Furthermore, few Pakistani studies use a **multi-cycle action research model** to track changes over time. This study contributes to the literature by implementing a structured intervention across **three cycles**, enabling continuous reflection and improvement. Finally, the integration of STEAM methodologies with Science and Social Studies textbooks remains underexplored. This research fills that gap by merging curriculum content with creative, practical, and interdisciplinary learning experiences.

2.7 Summary

The literature clearly establishes the urgency of addressing Pakistan's water crisis and the importance of school-based environmental education. Although various initiatives exist, major gaps persist in terms of student engagement, interdisciplinary teaching, critical thinking, and psychomotor skill development. This action research contributes to filling these

gaps by implementing a student-centered awareness campaign grounded in STEAM, Islamic values, and critical pedagogy. The study not only enhances students' knowledge but also fosters responsibility, creativity, leadership, and sustainable water-use habits—ultimately contributing to national and international research on water conservation education.

Chapter 03 Research Methodology

3.1 Research Design

This study adopts a qualitative action research design, focusing on exploring and understanding the impact of water conservation awareness campaigns on Grade 7 students' behaviors and attitudes towards sustainable water usage. Action research is an effective methodology in educational settings as it empowers teachers to investigate and improve their own practices while directly involving students in the learning process (Creswell, 2014). The cyclical nature of action research involving planning, action, observation, and reflection makes it particularly suitable for classroom-based interventions aimed at promoting environmental awareness and behavioral change.

A qualitative approach is appropriate for this study as it seeks to gain in-depth insights into students' perceptions, experiences, and evolving attitudes towards water conservation, rather than focusing on numerical data. Methods such as observation checklists, students' reflective journals, focus group discussions, and semi-structured interviews with teachers and students will be employed. These tools allow for a rich, descriptive understanding of how awareness campaigns influence students' daily water usage habits and their sense of environmental responsibility.

In the context of Pakistani schools, qualitative action research has been successfully employed to address environmental and educational challenges, fostering participatory learning and promoting behavioral change. For example, Bukhari and Kamal (2017) used qualitative methods

to examine students' environmental awareness in Punjab, highlighting the effectiveness of classroom-based interventions in cultivating pro-environmental attitudes. Similarly, Fayyaz et al. (2023) recommended qualitative research designs for environmental education projects in Karachi schools, emphasizing the importance of student-centered, experiential learning approaches in promoting conservation behaviors.

By utilizing a qualitative action research framework, this study not only investigates the effectiveness of awareness campaigns in enhancing water conservation habits but also contributes valuable insights for educators and policymakers seeking to integrate environmental education more meaningfully into the primary school curriculum in Pakistan.

3.2 Research Site and Participants

The research was conducted in Grade 7 classroom involving 6 students as a sample at, Ibrahim Ali Bhai Government Girls Secondary School FB, Area Karachi Pakistan.

3.3 Data Collection Tools

To comprehensively evaluate the impact of water conservation awareness campaigns on Grade 7 students, this study employed multiple qualitative data collection tools, each designed to capture different aspects of students' knowledge, attitudes, and behaviors. These are as follows:

- Semi-Structured Interview
- FGD (focused group discussion)
- Daily Diary Reflection
- Lesson Plans

3.4 Procedures of Data Collection (Action Plan)

Qualitative data was obtained from FGD, daily dairy reflection, and semi- structured interview, and analyze using thematic analysis to extract key insights and identify changes in knowledge and behavior.

3.4.1 Reconnaissance The preliminary stage

In this preliminary study, the researcher collected semi-structured interview from teacher to perform baseline assessment of water conservation awareness of grade 7 Science teacher at Ibrahim Ali Bhai Government Girls Secondary School.

3.4.2 Cycle One

Focused Group Discussion (FGD):

Teacher researcher conducted FGD to gain insights into participants prior knowledge, perceptions, and behaviors related to water conservation habits and their understanding of the essentialness of water in our lives. Additionally, through these FGDs, the teacher-researcher gained valuable insights into the instructional strategies used to teach water awareness topics in the participants classes.

Lesson Plan

The teacher-researcher used lesson plans in cycle one and subsequent cycles the teacher-researcher utilized lesson plans to implement the intervention for awareness campaign activities embrace students motivation, understanding and allowing them to became the active participants or learners in the classroom.

Daily Diary Reflection

The teacher-researcher used daily diary reflection in cycle one and subsequent cycles, after each lesson, time is taken to reflect on what worked well and what didn't. Thoughts, Observations, and insights are documented in a journal. This reflection helps to identify areas for improvement and informs planning for future lessons, refining the teaching approach and better supporting student learning.

3.4.3 Cycle Two

Lesson Plan

I employed lesson plans as a key tool to facilitate student learning and engagement. These lesson plans were designed to not only impart knowledge about water conservation but also to inspire critical thinking, empathy, and action among students by incorporating real-world example and interactive activities, I aimed to help students connect the dots between water scarcity, food security, environmental change, and human well-being. The lesson plan also encouraged students to generate their own suggestions and solutions for conserving water, fostering a sense of ownership and agency. This approach allowed me to assess the effectiveness of the awareness campaign in promoting water in promoting water conservation habits and inspiring students to become agents of change.

Daily Diary Reflection

The teacher-researcher used the insights of daily diary reflection to assess the improvement, and the readiness of students for moving them towards the advance level from basic understanding level by inculcating the concepts of social studies and science together as well to nurture students critical thinking / giving students voice to make them active learners by getting their suggestions

in serious issue like environmental change along with water scarcity or water drought , water pollution along with marine life effect and health issues.

3.4.5 Cycle Three

Lesson Plan

I utilized lesson plans as a tool to implement an awareness campaign on water conservation. The lesson plans consisted of interactive activities, including recycling bottle making and planting, aimed at raising awareness about the importance of water conservation, environmental awareness and sustainability. Students were divide into two groups “Water Watcher” and “Aquatic Ambassadors”. Water Watcher identified and reported leaking taps and dispensers, educated their peers on efficient water use. Both groups participated in planting activities, created recycled plastic bottle planters, and made posters promoting water conservation. Students presented their work, showcasing their posters, planters, and speech highlighting the importance of water saving for Pakistan.

Semi-Structured Interview

The teacher-researcher conducted semi-structured interview from the same science teacher of grade 7 student for knowing about the improvement teacher had felt in students after the awareness campaign.

Focused Group Discussion

The teacher-researcher gain the feedback / vision of participants after performing awareness campaign as well to compare the knowledge, behavior and perceptions about water essentialness, water conservation habits as well their affection / empathetic development towards this issues.

3.5 Ethical Considerations

This action research was placed a high premium on its ethical aspects. The relationship will be based on mutual trust and cooperation because action research fosters collaboration between the researcher and the participants. The school administration, was asked for their informed consent. All participant's identities were kept private to protect their privacy, and participation was completely voluntary. All stakeholders engaged received a clear communication about the goal of the study and its possible advantages.

Chapter 04: Data Analysis

The data for this action research were collected in three stages—pre-intervention, intervention, and post-intervention—and analyzed using thematic analysis. During the pre-intervention stage, data were gathered through a semi-structured interview with the Science teacher and a Focused Group Discussion (FGD) with six Grade 7 students at Ibrahim Ali Bhai Govt. Girls Secondary School, Karachi. These six students were selected through purposive sampling to ensure the inclusion of participants who could meaningfully contribute to the study. The intervention stage further required students with varying learning levels, enabling the collection of diverse viewpoints and providing a more accurate representation of class-wide behaviors.

During the intervention phase, data was collected across **three cycles**, where each cycle included lessons, daily diary reflections, and student activities linked to the Awareness Campaign on water conservation.

Finally, in the post-intervention stage, the impact of the awareness campaign on students' knowledge, perceptions, and water conservation behavior was assessed through a **post-FGD**, teacher semi-structured interviews, and daily reflection journals.

A thematic analysis revealed four key themes:

- Voices of Public- School People on the Water Crisis Issue
- Integrating Students' Experiences: Developing Water Habits through
- Awareness Campaign
- Awareness Campaign as a Teaching Method

- The Role of Awareness Campaign in Developing Water Conservation Habits in Grade 7 Students

4.2 Pre-Intervention

4.2.1 Voices of Public- School People on the Water Crisis Issue

During this stage, data was gathered through a semi-structured interview with the Science teacher and a Focus Group Discussion (FGD) with Grade 7 students. The aim was to explore existing knowledge, perceptions, and behaviors regarding water conservation, as well as observe current teaching practices in the school.

4.2.2 Teachers' Perceptions on Current Teaching Practices

The Science teacher revealed that water-related topics were primarily delivered through traditional textbook-based instruction, without the use of interactive or hands-on activities. She explained, *"I mostly teach from the textbook because we often have unexpected off days, and many students cannot afford materials for activities."* This indicated a reliance on conventional teaching methods and highlighted the structural challenges the teacher faced.

Furthermore, when I asked about her familiarity with global educational frameworks, she shared, *"I have heard about the SDGs, but I do not know much about how to use them in teaching."*

Although the school had previously participated in STEAM fairs with higher classes, she admitted that she had not incorporated STEAM approaches in her own lessons, stating, *"STEAM projects were mostly done by the upper classes; I never used STEAM activities with Grade 7."*

The teacher also mentioned including religious perspectives in her lessons, emphasizing moral responsibility in water usage. As she noted, *"I do tell the girls that wasting water is a sin and*

that we will be answerable to Allah.” However, while she embedded Islamic teachings effectively, she did not contextualize water scarcity within wider national or global issues, such as climate change, policy failures, or international water challenges.

Overall, the teacher’s insights highlighted important gaps in pedagogical practices, including limited exposure to experiential learning, the absence of STEAM-based instruction, and minimal integration of global environmental perspectives—gaps that this action research aimed to address through structured, student-centered awareness campaign cycles.

4.2.3 Students’ Perceptions, Knowledge, and Behavior (DROPS Framework)

To explore Grade 7 students’ understanding and behaviors related to water usage, the DROPS Framework (Driving Realistic Outcomes for Planet Sustainability) was applied during the pre-intervention FGD. The insights revealed clear gaps in knowledge, inconsistencies in perceptions of classroom teaching, and limited awareness of water conservation practices.

✓ DROPS-K: Knowledge Gaps

Students demonstrated only a basic understanding of water-related topics. While they could identify common sources of water—such as rivers, lakes, rain, and tube wells—their conceptual knowledge was limited. All six participants reported unfamiliarity with the Water Cycle, stating that they had neither studied nor understood it in English or Urdu. Similarly, most students showed limited awareness of environmental concepts such as water pollution, drought, and flooding; only one student referred to the recent floods in Punjab, suggesting that exposure to current events influenced individual understanding.

✓ **DROPS-P: Perceptions of Current Teaching Practices**

A notable insight emerged regarding students' perceptions of how Science was taught in their classroom. Four students openly shared that the teacher relied primarily on textbook reading and teacher explanation, with little to no use of activities, posters, or real-life examples. Their responses aligned with the teacher's own statement about depending on the textbook due to limited time and resources.

However, two students insisted that their teacher regularly conducted activities, used posters, and engaged the class in real-life discussions. During the FGD, these two students repeatedly encouraged others to agree with them, saying things like, "Say yes, our teacher does activities." This attempted influence suggested a sense of hesitation or fear, possibly stemming from the belief that their responses might affect their relationship with the teacher. Their insistence contrasted sharply with both the teacher's interview and the observations shared during the discussion. Their insistence contrasted sharply with both the teacher's interview and the majority of student responses, revealing an important social dynamic influencing student honesty.

This difference highlighted the power dynamics present in classroom-based research and reinforced the importance of establishing trust and reassurance during data collection

✓ **DROPS-P: Preferred Teaching Methods**

Despite the variation in their reported experiences, all six students unanimously expressed a desire for more engaging and interactive learning approaches. They specifically mentioned wanting:

- games
- hands-on activities

- stories
- poster-making
- real-life examples

This collective preference clearly indicated an unmet need for student-centered instructional methods that could support deeper understanding and retention.

✓ **DROPS-B: Behavior Toward Water Conservation**

Students reported practicing only basic water-saving behaviors. Some mentioned turning off taps after use or conserving water during wudu. However, most students admitted they were not consistent in applying these habits and had never actively reflected on their personal water usage.

Their responses showed that although they possessed a general awareness of water conservation as a “good action,” they lacked the deeper understanding, motivation, and structured guidance needed to form long-term habits.

Pre-Intervention Baseline Summary

The data from the pre-intervention stage clearly established that:

- Students had limited conceptual knowledge about water conservation and environmental issues.
- Teaching practices were predominantly traditional, reinforcing passive learning.
- A social element of peer influence affected the authenticity of some student responses.
- Students desired more interactive and meaningful learning experiences.
- Water-saving behaviors existed but were superficial and inconsistent.

These findings provided a strong foundation for designing an intervention focused on **critical thinking, active participation, student leadership, and real-world application**, all were embedded into awareness campaign cycles.

4.3 Intervention (Cycles)

4.3.1 Integrating Students' Experiences: Developing Water Habits through Awareness Campaign

Across three cycles of action research, lessons were designed to build **knowledge**, strengthen **critical thinking**, and lead students toward **real-life application** of water conservation practices. Each cycle was supported by lesson plans and daily diary reflections.

Theme 02 Integrating Students' Experiences: Developing Water Habits through Awareness Campaign

This Awareness Campaign, grounded in Paulo Freire's critical pedagogy, guides students' through Bloom's Taxonomy, from remembering key concepts to creating innovative solutions for water conservation, by fostering conscientization, critical thinking, and praxis, Awareness campaign empower students to become an agent of change, while contributing to Quality Education (SDG 4), Environmental Awareness (SDG 13-15), and Life Below Land (SDG 15).

4.1.1 Cycle One (Knowledge Building)

Two lessons were taught in this cycle, supported by daily reflection.

Lesson 01 Sources of Water and the Water Cycle

The first lesson introduced students to the **distribution of fresh and salty water on Earth**, helping them understand how limited fresh water actually is. Students explored water sources through the lively "**Catch the Ball**" activity, where each student who caught the ball had to

name a water source such as rivers, lakes, oceans, or groundwater. This playful activity immediately boosted participation and attention.

To promote responsible water use, the teacher introduced the slogan

“Water! Water! Everywhere — Keep it clean, use it with care.”

This simple message helped students understand that conserving water is needed not just globally but in their everyday life.

Students then learned about the **water cycle** using visual aids and familiar examples. To reinforce the concept, a **“Water Cycle Fast Drawing Challenge”** was held, where selected group representatives raced to draw and label the water cycle on the board. Meanwhile, their group members also completed the same activity at their desks. This activity encouraged teamwork, creativity, and quick recall.

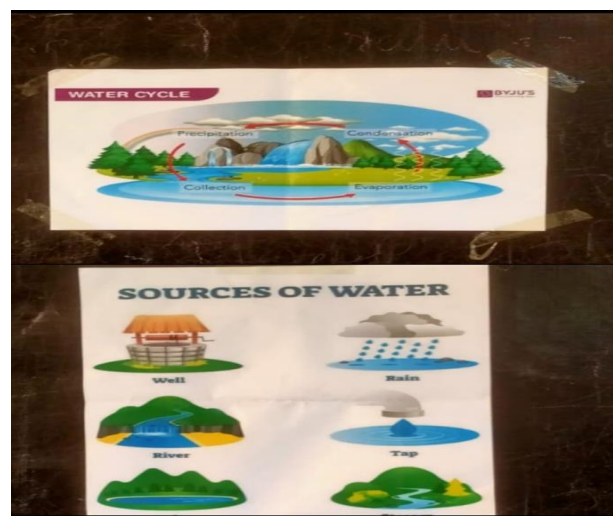


Figure 1 sources of water

4.1.2 Result of Lesson plan 01

The lesson successfully introduced students to the importance of water conservation and the water cycle. Through interactive activities, students gained a deeper understanding of the

distribution of water on Earth, sources of water, and the water cycle process, as well the slogan promoted awareness and encouraged responsible water usage. Students demonstrated increased knowledge and appreciation for water conservation, setting a strong foundation for future learning.

4.1.4 Result of Daily Diary Reflection

The teacher-researcher utilized daily diary reflection. This involved teacher-researcher observation after every lesson, it includes targeted students responses, their interest building and the understanding about the topics and their enhancement of critical thinking.

Lesson 02 River and People

To make the topic engaging, the teacher entered the classroom dressed as a river, creatively symbolizing the life and importance of rivers. This surprising introduction immediately captured students' attention and helped them emotionally connect with the topic.

Students learned the names of the major rivers of Pakistan through a simple mnemonic trick, and they participated in a Charades Game, acting out river-based occupations such as fishermen, farmers, and boatmen. This helped them understand how many people depend on rivers for their livelihood.

Science concepts were integrated by discussing river systems upper, middle, and lower courses and the effects of water pollution on the environment. Social Studies concepts were also incorporated, including how human activities affect river ecosystems and why rivers hold cultural and economic importance.

Before ending the lesson, students wrote suggestions for saving rivers and pasted them on a class chart, taking their first step toward becoming environmental stewards.

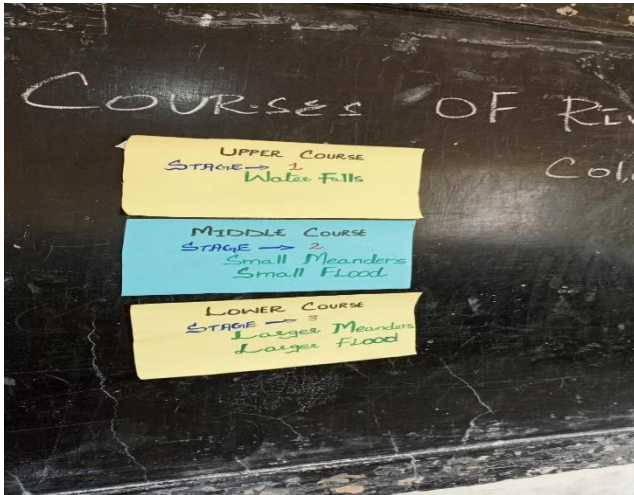


Figure 2 Courses of river



Figure 3 Ways to save river



Figure 4 Environmental awareness

Result of Lesson Plan

The lesson plan on River and People introduced students to river courses, Occupations, and water pollution impacts. Students understood the importance of river conservation and suggested ways to save river water. This aligns with SDGs Clean Water and Sanitation (6), Climate Action

(13), Life below Water (14), and Life on Land (15). The awareness campaign developed empathy towards rivers, marine life as well life on land which is getting affected as well promotes critical thinking and problem-solving skills, laying a foundation for advance concepts in water conservation.

Result of Daily Diary Reflection

The teacher-researcher utilized daily diary reflection. This involved teacher-researcher observation after every lesson, it includes targeted students responses, their interest building and the understanding about the topics and their enhancement of critical thinking allow teacher-researcher to move students forward from basic level of knowledge to the advance level.

Connection to Cycle 2 Advance Learning

Building on the foundation established in Cycle 01, Cycle 02 was delved deeper into water pollution, floods, water droughts, and irrigation system in Pakistan.

Students would explore the causes and effects of water pollution and other issues mentioned above and discuss ways to prevent it.

The knowledge gained in Cycle 01 would serve as a precursor to understanding the importance of water conservation and management in Cycle 02.

4.1.4 Cycle Two (Advance Learning)

In the second cycle, the teacher-researcher use Lesson Plan to introduce the idea of integrated lessons of Science and Social Studies concepts on Environmental awareness and Water issues. and daily diary reflection to observe the difference in students' attitude, knowledge and perspective after the initial cycle, additionally teacher-researcher observe the effectiveness of awareness campaign on students.

Lesson Plan 03 Irrigation System in Pakistan

This lesson began by revisiting what students had learned earlier about the importance of rivers to people's daily lives. Building on this prior knowledge, students explored Pakistan's irrigation system and how much the country depends on rivers for agriculture and food security.

To deepen their moral and ethical understanding, students learned an important Hadith of Prophet Muhammad (PBUH):

“Do not waste water, even if you are at a flowing river.”

This Hadith helped students realize that water conservation is not only a scientific responsibility but also an Islamic duty.

Science concepts such as water management and environmental impact were explained, while Social Studies concepts included the role of irrigation in communities and economies. Students were encouraged to think critically about water waste, the challenges of supplying water to farmers, and the need for sustainable practices.



Figure 5 irrigation system



Figure 6 Pakistan irrigation system

Result of Lesson Plan

This lesson fostered critical thinking, ethical values, and awareness about water conservation, connecting Islamic perceptions with real-life issues. It enhanced understanding of irrigation systems, emphasized responsible water management, and highlighted its impact on communities and the environment, ultimately improving knowledge and promoting sustainability. The lesson also nurtured students' affective domain, instilling values like gratitude, responsibility, and compassion, as emphasized in Islamic teachings, encouraging them to appreciate water as a blessing and use it wisely.

Result of Daily Diary Reflection

The daily dairy reflection serves as a valuable tool for the teacher-researcher to assess the progress of student's participation, knowledge, and behavioral changes from the first cycle to the second cycle of the third lesson. This reflection helps evaluate the fulfilment of blooms taxonomy across cognitive and affective domain. It indicates significant improvement in

student's intellectual thoughts and thinking, preparing them to advance to the third cycle with enhanced response and engagement.

Developed a sense of water conservation towards butterfly effect

Lesson 04 Water pollution, Water Drought and Floods

This lesson addressed some of Pakistan's most pressing environmental challenges. Students learned the causes and effects of water pollution, droughts, and floods through an interdisciplinary approach combining Science and Social Studies.

Students shared their insights about human activities that cause water pollution, and the teacher built upon their responses with further explanation. Connecting the topic to real-life experiences, the teacher discussed the 2022–2023 floods in Pakistan, helping students understand how environmental disasters affect health, the economy, animals, and communities.

Students felt emotional empathy for the victims and expressed a desire to create change. They began brainstorming ideas such as recycling, planting trees, and practicing water conservation.

In a hands-on activity, students used recycled plastic bottles as plant holder pots, taking their first real step toward environmental action. This represented a Butterfly Effect, showing how small steps can lead to larger impact.



Fig 7 Water scarcity impact



Fig 8 Students recycling plastic bottles



Fig 9 Student's planters

Result of Lesson Plan

This lesson plan helps students become environmentally conscious, equipped with critical thinking and empathy. Their actions can spark a Butterfly Effect, inspiring broader change.

4.1.5 Result of Daily Diary Reflection

The result from the second cycle showed up that students became more concerned about water conservation, environmental awareness, and understood about the butterfly effect. Moreover, they understood that saving water is not only essential for preventing the crises as well it's an act of showing gratitude to Allah, this awareness campaign broadened their horizons, enabling them

to see issues from a more nuanced perspective, and they began to search countries suffering from water scarcity to prevent our own country Pakistan from the severe crises of water and environmental change impacts. As well awareness campaign makes students' an agent of change.

4.1.6 Cycle Three (Action and Application)

In the third cycle, teacher researcher employed three tools. Initially, Lesson Plan, focused group discussion (FGD) were conducted using open-ended question to gather insights. Next, semi-structured interview were held with targeted Science and Social Studies teachers to gain their perspective on the effectiveness of Water Awareness Campaign in enhancing students' learning. The aim was to understand the perceived benefits of this campaign in promoting critical thinking about severity of water crisis. Lastly, daily diary reflections were used to evaluate the effectiveness of Bloom Taxonomy's cognitive domain levels, starting from remembering, moving towards understanding and application, and ultimately reaching the level of creation, thereby covering both lower- and higher- order thinking.

Lesson Plan 05 Creating Awareness Materials

In this cycle, students transformed from learners into leaders. They were assigned roles as Water Watchers and Aquatic Ambassadors and began creating posters, slogans, and awareness materials promoting water conservation. They planted trees in the school garden, took part in campaigns, and involved their parents in designing creative posters.

This activity helped students apply what they had learned in earlier cycles and strengthened their teamwork, communication, and creativity.



Figure 10 Students learning planting from gardener



Figure 11 Students planted their plants



Figure 12 Students preparing posters

Result of Lesson Plan

The lesson has empowered students to take action and apply their knowledge of water conservation. They have developed important skills such as teamwork, communication, and problem-solving, and have demonstrated a commitment to making a positive Impact on their community.

Result of Daily Diary Reflection

The daily diary reflection shows that students were highly engaged in creating awareness materials and demonstrated a good understanding of water conservation concepts. Parental involvement was also notable, with many assisting students with poster making and providing feedback, as well encouraging their children towards their efforts and appreciating them a lot. This collaboration helped students develop their creativity and communication skills.

Lesson Plan 06 (Presenting and Sharing)

In the final lesson, students proudly showcased their speeches, posters, and awareness materials. They shared their reflections in front of classmates and teachers, demonstrating tremendous confidence and leadership. Their work was also shared in the class WhatsApp group, where parents expressed appreciation and pride.

Parents helped students prepare their speeches at home, strengthening their public speaking skills and reinforcing the importance of water conservation.



Figure 14 Students educating their peers



Figure 13 Students showcasing their work and presentation



Figure 15 Water Watchers performing their activity

Post- Intervention

Result of Lesson

The lesson was highly effective in empowering students to take action and apply their knowledge of water conservation. Students demonstrated significant improvement in their confidence and public speaking skills, and their understanding of water conservation has deepened. The parental involvement and support played a crucial role in reinforcing the importance of water conservation and encouraging students to take action. The use of STEAM approach and hands-on activities made the learning experience engaging and enjoyable for the students. This approach aligns with Paulo Freire's critical pedagogy theory, which emphasizes empowering students to become active agents of change.

4.1.7 Result of Focused Group Discussion (F G.D)

The final focus group discussion yielded similar themes to the pre-intervention stage. To gather insights, I used the same tool Focus group discussion- involving interviews with participants to assess their modified knowledge, behavior, and attitude towards water conservation habits.

Key Findings

The final stage of the intervention revealed a remarkable transformation in students' awareness, attitudes, and habits regarding water conservation. During the focused group discussion, students demonstrated a strong sense of empowerment, expressing that they now felt responsible for raising their voices about the water crisis facing Pakistan. One student shared, "Miss, we are aware now of the intensity of the water crisis, and we will take initiatives to bring change and save our country, In sha Allah." Such statements reflected a clear shift from limited understanding to informed action. The initial code DROPS-K3, which had earlier indicated a limited understanding of water-related concepts, showed significant improvement by the end of the campaign. Students began to demonstrate deeper conceptual knowledge, integrating scientific, social, and environmental aspects learned throughout the lessons.

As the discussion continued, students expressed a growing sense of pride in their contribution to water conservation. They enthusiastically stated, "Miss, we are feeling very proud and honored, and even our parents and teachers are appreciating us for our initiatives." Their responses illustrated not only cognitive growth but also a strong affective connection to the cause. Many students also reported applying new water-saving habits within their homes. One student reflected, "Miss, now I know many daily-life water-saving habits, and I practice them in my home and in school." This improvement corresponded to the initial code DROPS-B2, which had indicated limited application of water conservation practices in daily life. The post-intervention responses clearly showed visible behavioral change and commitment to sustainable habits.

The semi-structured interviews conducted with targeted teachers further supported these findings. Teachers reported a noticeable shift from traditional lesson delivery to a more participatory, student-centered approach as a result of the awareness campaign. They observed

that students were increasingly engaged, actively participating in discussions, and demonstrating enhanced critical thinking skills. Teachers expressed that students were not only more knowledgeable but also more confident in analyzing environmental issues, offering solutions, and relating classroom learning to real-world challenges.

Daily diary reflections from the teacher-researcher also indicated consistent growth across all three cycles. These reflections highlighted substantial improvement in students' logical reasoning and depth of understanding. The diary entries showed that as the campaign progressed, students began responding with greater clarity, coherence, and analytical ability. Parental involvement emerged as a particularly strong supportive factor, especially during activities such as recycling plastic bottles and creating posters. Parents provided constructive feedback to their children, which significantly boosted their self-esteem and enthusiasm for learning. This family-school collaboration contributed greatly to the development of students' confidence, creativity, and motivation to construct knowledge meaningfully.

Collectively, these findings demonstrate that the awareness campaign was not only effective in enhancing students' cognitive understanding but also successful in fostering emotional engagement and practical behavior change. Students transitioned from passive learners to active agents of change, capable of applying water-conservation habits in their daily lives and inspiring others within their communities. This progression strongly aligns with the principles of action research, highlighting reflective learning, continuous improvement, and meaningful transformation.

Reflection

Theme 03 Awareness Campaign as a teaching method

As a teacher-researcher, the overall findings of my action research clearly show that the awareness campaign had a profound impact on developing water conservation habits among my Grade 7 students. Throughout the intervention, I grounded my teaching approach in Paulo Freire's theory of critical pedagogy, which allowed me to shift students from passive learners to active participants who questioned, explored, and engaged meaningfully with real-world issues. All instructional content was directly linked with the Sindh Textbook curriculum, specifically Science Unit 4: "Water" and Social Studies Unit 05: "Water Resources," ensuring that my intervention remained curriculum-aligned while expanding students' conceptual depth.

In the cognitive domain, I observed a remarkable improvement in students' knowledge and higher-order thinking skills. At the beginning, their understanding of water-related concepts was limited, but as I taught them through STEAM-integrated lessons—covering the water cycle, water distribution, irrigation systems, floods, droughts, and pollution—their comprehension deepened. They moved beyond simple recall to demonstrating understanding, application, and even critical analysis. The activities I designed, such as fast-drawing challenges, river role-play, recycling tasks, and discussions on Pakistan's floods, helped make the Sindh Textbook content more meaningful and accessible.

The affective domain also showed significant growth. By integrating Islamic perspectives on water conservation, including the Hadith: *"Do not waste water, even if you are at a flowing river,"* I helped students connect environmental responsibility with their personal faith and values. As I collected their reflections and observed their participation, I noticed that they became more empathetic and emotionally invested in protecting water resources. Students

expressed genuine concern for communities affected by floods and droughts, and they took pride in adopting water-saving habits at home and school. This shift in their attitudes and sense of responsibility was evident in their comments, expressions, and voluntary participation.

In the psychomotor domain, students actively engaged in hands-on activities that strengthened their skills beyond content knowledge. Through poster making, planting activities, recycling bottles, and presenting their awareness messages, they developed communication, creativity, teamwork, and leadership. It was encouraging to see them practically applying the concepts I taught, transforming the classroom into a space of creation rather than memorization. Many students took ownership of the awareness campaign, confidently sharing their work with parents, peers, and teachers.

Overall, the data collected through focused group discussions, daily diary reflections, classroom observations, and semi-structured interviews reflected a clear progression across all domains of Bloom's Taxonomy. I witnessed a visible transformation—from basic understanding to meaningful application and creation. The integration of STEAM strategies, Islamic teachings, and Sindh Textbook content enabled my students to develop a deep understanding of water conservation while also empowering them to become active agents of change within their homes, school, and community.

Chapter 05 Discussion, Conclusion, and Recommendation

This chapter presents a detailed discussion of the findings in relation to previous studies. It also highlights how my research contributes to existing literature and classroom practices. The chapter concludes with implications, recommendations, limitations, and suggestions for future research.

5.1 Discussion

As a teacher–researcher, this action research allowed me to closely observe how an awareness campaign could influence the water conservation habits of seventh-grade students. Through three cycles of planning, action, observation, and reflection, I witnessed meaningful changes not only in students' knowledge but also in their attitudes and behaviors.

The findings clearly demonstrate that an awareness campaign—when combined with interactive activities, community involvement, and cross-disciplinary learning (STEAM and Social Studies)—can significantly strengthen students' understanding of water conservation.

Throughout the cycles, students gradually became more mindful about the importance of saving water and the consequences of wasteful habits.

Drawing on Paulo Freire's critical pedagogy, which emphasizes empowering learners to become active agents of change (Fayaz et al., 2023), I observed how students began taking ownership of the issue. Instead of seeing water conservation as just another school lesson, they treated it as a personal and community responsibility. This shift aligned strongly with Freire's notion of learner empowerment.

The “Butterfly Effect” was also visible in this research. I noticed that even small actions—such as reminding classmates not to waste water, creating posters, or discussing the topic at home—created a ripple effect within the school and surrounding community. These small but meaningful actions collectively generated a positive influence that extended beyond the classroom.

Additionally, integrating Islamic values related to water preservation strengthened the moral and ethical foundation of the campaign. Students learned that conserving water is not only a scientific and social responsibility but also an act of faith and gratitude. This connection made the learning experience more relevant and heartfelt for them.

Across the three action cycles, several encouraging patterns emerged:

- **Students' knowledge and enthusiasm increased noticeably.**

They became more engaged, asked thoughtful questions, and showed curiosity in understanding how their actions affected the environment.

- **Teachers adopted more innovative teaching strategies.**

My colleagues and I began exploring hands-on methods, inquiry-based activities, and collaborative tasks that enhanced the overall learning experience.

- **Community participation strengthened the campaign.**

Parents expressed appreciation, supported students' efforts at home, and even shared examples of water conservation practices they started implementing.

These findings align with previous literature highlighting the positive impact of awareness campaigns in educational settings. As noted by Fayaz et al. (2023), such campaigns can bring

about meaningful behavioral changes when students are given opportunities to understand real-world issues through engaging and collaborative learning experiences.

Key Takeaways

- Awareness campaigns can significantly influence environmental behaviors (Fayaz et al., 2023).
- Integrated and interactive approaches promote deeper understanding and sustained actions.
- Students can serve as powerful agents of change within their schools, homes, and communities.

By utilizing awareness campaigns, educators can help nurture environmentally responsible future citizens capable of addressing global challenges such as water scarcity with confidence and compassion.

5.2 Conclusion

In conclusion, this action research confirms that awareness campaigns are effective tools for promoting water conservation habits among seventh-grade students. Through the integration of interactive activities and cross-disciplinary strategies, students developed a more meaningful and personal understanding of water conservation.

As a teacher-researcher, I observed noticeable changes in the way students thought about and used water. They not only learned new information but also translated that knowledge into

action. Many of them confidently participated in discussions, initiated projects, and motivated peers to adopt more responsible behaviors.

The findings reinforce the idea that awareness campaigns—when combined with student empowerment—can lead to lasting behavioral change. Students are capable of becoming influential advocates for environmental protection when given the right guidance, opportunities, and encouragement.

5.3 Implications

The implications of this study are significant for educators, policymakers, and educational institutions:

- **For teacher training:**

Awareness campaigns and STEAM-based activities should be included in professional development programs. This will prepare teachers to incorporate environmental topics in engaging and meaningful ways.

- **For curriculum development:**

Schools should revise curricula to include more opportunities for environmental education. Integrating awareness campaigns can help foster a culture of sustainability from an early age.

- **For policymakers:**

There is a need to recognize environmental education as an essential component of responsible citizenship. Policies should encourage schools to adopt programs that promote conservation practices.

- **For educational institutions:**

By creating an inclusive, supportive learning environment, schools can help students develop habits that positively influence their communities.

When educators and policymakers work together, they can empower students to adopt sustainable habits and become agents of positive change.

5.4 Recommendations

Based on my findings, I recommend the following:

1. **Integrated Environmental Education:**

Include water conservation topics across different subjects to encourage responsible habits from a young age.

2. **Interactive Teaching Methods:**

Engage students with hands-on activities, games, projects, and community involvement to deepen their understanding.

3. **Community and Parental Involvement:**

Encourage families and communities to participate in school campaigns so that learning extends beyond the classroom.

4. **Student Leadership Opportunities:**

Provide platforms for students to take the lead in environmental initiatives, helping them develop confidence and leadership skills.

5.5 Limitations

While the study produced meaningful results, several limitations must be acknowledged:

- **Sample Size:**

The study focused only on seventh-grade students, which limits the generalizability to other age groups or larger populations.

- **Single-Gender Group:**

The participants were all female students, which may affect the applicability of the findings to mixed-gender or male groups.

- **Contextual Factors:**

School location, cultural norms, and community characteristics may have influenced the results.

- **Timeframe:**

The duration of the action research may not have been long enough to measure long-term behavioral changes.

5.6 Future Research

To build on this study, future researchers may consider:

- **Longitudinal Studies:**

To observe the long-term effects of awareness campaigns on students' environmental behaviors.

- **Diverse Populations:**

Including different age groups, genders, socioeconomic backgrounds, and school environments.

- **Comparative Studies:**

Comparing various awareness campaign models to identify the most effective strategies.

- **Scaling Up Initiatives:**

Exploring ways to expand successful campaigns to larger populations for broader impact.

By addressing these areas, future research can help strengthen strategies for promoting environmental awareness and contribute to a more sustainable world.

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Appendix-A

Lesson Plan 01

Lesson Plan: Sources of Water and water cycle

Class: 7

Subject: Science/social studies

Time: 45 minutes

Objectives

1. Students will be able to identify and describe different sources of water.
2. Students will understand the water cycle process.
3. Students will participate in activities that promote critical thinking and creativity.

Materials

- Blackboard and colorful chalks
- Diagram of the water cycle
- Ball
- Paper and pencils for drawing

Development

1. Introduction (5 minutes)

Begin by asking students about the importance of water and where it comes from.

Write their responses on the board and introduce the concept of sources of water (e.g., rivers, lakes, oceans, groundwater, rainwater).

2. Catch the Ball and Name the Source (10 minutes)

Throw the ball to a student and ask them to name a source of water.

The student then throws the ball to another student, who names a different source of water.

Continue until all students have participated.

3. Water Cycle Process (10 minutes)

Explain the water cycle process using a diagram:

1. **Evaporation** Water evaporates from the Earth's surface into the air.

2. **Condensation** Water vapor condenses into clouds.

3. **Precipitation** Water falls back to the Earth as rain, snow, or hail.

4. **Runoff** Water flows over the ground and into bodies of water.

Use simple examples to illustrate each stage.

4. Fast Drawing of Water Cycle Competition (15 minutes)

Divide the class into small groups and give each group paper and pencils.

Set a timer for 3 minutes and ask students to draw the water cycle process.

Encourage creativity and accuracy.

The group with the best drawing wins.

Assessment

Observe student participation during activities.

Review their drawings of the water cycle process.

Ask questions throughout the lesson to assess understanding.

Lesson Plan 02

Lesson Plan: River and People

Class: 7

Subject Social Studies/Science

Time 45 minutes

Objectives

1. Students will identify the different courses of a river.
2. Students will learn about the benefits of rivers to people and the environment.
3. Students will recognize how human activities impact rivers and marine life.
4. Students will think critically about ways to conserve and save rivers.

Materials

- Blackboard and chalks
- Diagram of a river and its courses
- Guest speaker (fellow teacher) dressed as a river
- Chart paper
- Sticky notes or stickers
- Markers or colored pencils

Procedure

1. Introduction (5 minutes)

Introduce the concept of a river and its different courses (upper, middle, and lower).

Show a diagram of a river and explain the characteristics of each course.

2. Teacher portrayed as River Ravi (10 minutes)

teacher dressed as a river enters the classroom and introduces themselves as "Ravi" (or any other river name).

Ravi explains the benefits of rivers to people, such as:

Providing water for drinking, irrigation, and industry

Supporting aquatic life and biodiversity

Facilitating transportation and trade

Ravi also explains how people harm rivers, such as:

Pollution from industrial and agricultural waste

Over-extraction of water for human use

Destruction of habitats and ecosystems

3. Upper Course (5 minutes)

Ravi explains the characteristics of the upper course, such as:

Fast-flowing water

Rocky terrain

Supports unique plant and animal species

Discuss occupations related to the upper course, such as:

Farming and forestry

Mining and quarrying

4. **Middle Course** (5 minutes)

Ravi explains the characteristics of the middle course, such as:

Slower water flow

Wider river channel

Supports agriculture and fishing

Discuss occupations related to the middle course, such as:

Farming and irrigation

Fishing and aquaculture

5. **Lower Course** (5 minutes)

Ravi explains the characteristics of the lower course, such as:

Slow-moving water

Wide river mouth

Supports shipping and trade

Discuss occupations related to the lower course, such as:

Shipping and transportation

Fishing and seafood harvesting

6. **Environmental Impact** (5 minutes)

Discuss how human activities impact rivers and marine life, such as:

Pollution and climate change

Overfishing and habitat destruction

Invasive species and disease

7. Activity River Conservation Suggestions (10 minutes)

Distribute sticky notes or stickers and markers or colored pencils to students.

Ask students to write or draw their suggestions for conserving and saving rivers.

Have students stick their suggestions on a chart paper titled "Save Our Rivers".

Encourage students to be creative and think critically about ways to protect rivers.

Assessment

Observe student participation during the lesson.

Ask questions throughout the lesson to assess understanding.

Review student suggestions on the chart paper for understanding and creativity.

Lesson Plan 03

Lesson Plan: Irrigation System of Pakistan

Class 7

Subject Social Studies/Environmental Science

Time 45 minutes

Objectives

1. Students will learn about the irrigation system of Pakistan and its importance.
2. Students will recognize the impact of irrigation on the environment and rivers.
3. Students will relate the Islamic teachings on water conservation.

Materials

- Blackboard and chalks
- Diagram of Pakistan's irrigation system
- Pictures of irrigation canals and dams

Procedure

1. Introduction (5 minutes)

Ask students what they know about irrigation and its importance.

Introduce the topic of irrigation system of Pakistan.

2. Irrigation System of Pakistan (10 minutes)

Explain the irrigation system of Pakistan, including:

- Canals

- Dams
- Barrages

Show a diagram and pictures of irrigation canals and dams.

Discuss the benefits of irrigation, such as:

- Growing crops

Providing water for drinking and other uses

3. Impact on Environment and Rivers (10 minutes)

Discuss the impact of irrigation on the environment and rivers, such as:

- Waterlogging and salinization
- Water scarcity
- Pollution

Explain the importance of using water wisely.

4. Islamic Teachings on Water Conservation (10 minutes)

Explain the importance of water conservation in Islam.

Quote the Hadith of Prophet Muhammad (PBUH):

"Do not waste water even if you are at a flowing river."

Discuss how we can conserve water in our daily lives.

5. Activity (5 minutes)

Ask students to write one way they can conserve water at home.

Students share their answers with the class.

6. Conclusion (5 minutes)

Summarize the importance of irrigation system of Pakistan and the need for water conservation.

Assessment

Observe student participation during the lesson.

Ask questions throughout the lesson to assess understanding.

Review student answers in the activity.

Lesson Plan 04

Lesson Plan: Water Pollution, Droughts, and Floods

Class 7

Subject Environmental Science/Social Studies

Time 2 units (90 minutes)

Unit 1 Water Pollution, Droughts, and Floods (45 minutes)

Objectives

1. Students will identify the causes and effects of water pollution, droughts, and floods.
2. Students will learn about the impact of these issues on living beings and the environment.
3. Students will develop empathy and take initiative to make a positive change.

Materials

- Blackboard and chalks
- Pictures and videos of water pollution, droughts, and floods
- Diagrams of the water cycle and the effects of pollution
- Handouts with statistics and real-life examples

Procedure

1. Introduction (5 minutes)

Ask students what they know about water pollution, droughts, and floods.

Introduce the topics and explain their importance.

2. **Water Pollution** (10 minutes)

Explain the causes of water pollution, such as

- Industrial waste
- Agricultural runoff
- Domestic waste

Discuss the effects of water pollution, such as

- Harm to marine life
- Human health problems
- Economic impacts
- Show pictures and videos of water pollution.

3. **Droughts and Floods** (10 minutes)

Explain the causes of droughts and floods, such as

- Climate change
- Deforestation
- Poor water management

Discuss the effects of droughts and floods, such as

- Impact on agriculture and food security
- Displacement and migration of people
- Loss of life and property
- Show pictures and videos of droughts and floods.

4. Real-Life Connectivity (10 minutes)

Discuss the 2022-2023 floods in Pakistan and their impact on

Human life and health

Economy and infrastructure

Environment and wildlife

Explain how these issues are connected to water pollution, droughts, and floods.

5. Activity (10 minutes)

Ask students to reflect on what they have learned and how they can make a positive change.

Encourage students to share their thoughts and ideas.

2. Unit : Taking Action (45 minutes)

Objectives

1. Students will learn about the importance of recycling and conservation.
2. Students will create recycled plastic bottle planters.

Materials

- Plastic bottles
- Scissors
- Soil
- Seeds
- Markers or paint

Procedure

1. Introduction (5 minutes)

Explain the importance of recycling and conservation.

Introduce the activity: creating recycled plastic bottle planters.

2. Creating Planters (30 minutes)

Have students cut the top off plastic bottles and decorate them with markers or paint.

Fill the bottles with soil and plant seeds.

Have students write their names and the name of the plant on the bottle.

3. Conclusion (10 minutes)

Students will share their planters and discuss the importance of taking small steps to make a positive change.

Students will take their planters home and make them complete if left.

Assessment

Observe student participation during the lesson.

Ask questions throughout the lesson to assess understanding.

Review student planters for creativity and understanding.

Lesson Plan 05

Lesson Plan: Creating an Awareness Campaign on Water Conservation

Class 7

Subject Environmental Science/Social Studies

Time 45 minutes

Objectives

1. Students will learn the importance of water conservation and its impact on the environment.
2. Students will recognize the role of awareness campaigns in promoting environmental conservation.
3. Students will create awareness materials and participate in a planting activity to promote water conservation.

Materials

- Chart paper
- Markers
- Posters
- Scissors
- Glue
- Plants
- Soil
- Watering cans

Procedure

1. Introduction (5 minutes)

Teacher will ask students what they know about water conservation and its importance.

Introduce the concept of awareness campaigns and their role in promoting environmental conservation.

2. Group Discussion (10 minutes)

Divide students into small groups and ask them to discuss the following questions

Why is water conservation important?

How can we conserve water in our daily lives?

What are the consequences of water scarcity?

Each group share their ideas with the class.

3. Creating Awareness Materials (15 minutes)

Provide students with chart paper, markers, scissors, and glue.

Ask students to create posters and banners that promote water conservation.

Encourage students to be creative and include catchy slogans and images.

4. Planting Activity (10 minutes)

Take students to a designated area in the school for a planting activity.

Students plant trees or flowers and water them.

Students will take care of the plants and monitor their growth.

5. Conclusion (5 minutes)

Ask students to share what they have learned and how they plan to promote water conservation in their daily lives.

Assessment

Observe student participation during the lesson.

Review student posters and banners for creativity and understanding.

Ask questions throughout the lesson to assess understanding.

Lesson Plan 06

Lesson Plan: Presenting and Sharing - Water Conservation Awareness

Class 7

Subject Environmental Science/Social Studies

Time 45 minutes

Objectives

1. Students will develop public speaking skills and confidence.
2. Students will learn the importance of water conservation and its impact on the environment.
3. Students will educate their peers on the essentialness of water and the importance of conserving water.

Materials

- Posters and banners created by students with parental help
- Water conservation materials (handouts, brochures, etc.)

Procedure

1. Introduction (5 minutes)

Teacher will welcome students and introduce the presenters, "Water Watchers" and "Aquatic Ambassadors".

Explain the purpose of the presentation: to educate peers on the importance of water conservation.

Acknowledge the support of parents in helping students prepare for this presentation.

2. Presentations (20 minutes)

Group 1 "Water Watchers"

Presenters will share their research and findings on the essentialness of water.

Showcase posters and banners highlighting the importance of water conservation, created with parental help.

Group 2 "Aquatic Ambassadors"

Presenters share their research and findings on the impact of water pollution and waste.

Showcase posters and banners highlighting ways to conserve water, created with parental help.

3. Q&A Session (10 minutes)

Open the floor for questions and answers.

Encourage peer-to-peer discussion and learning.

4. Call to Action (5 minutes)

Presenters emphasize the importance of conserving water and encourage peers to take action.

Distribute water conservation materials (handouts, brochures, etc.).

5. Conclusion (5 minutes)

Thank presenters, students, and parents for their participation and support.

Emphasize the empowerment of students as change-makers in their community.

Assessment

Observe student participation and engagement during the presentation.

Evaluate student understanding and retention of water conservation concepts.

Review posters and banners for creativity and effectiveness.

Appendix-B

Teacher Interview Questions

1. How do you usually teach topics related to water, rivers, and the water cycle in your class?
2. Do you include activities (like experiments, posters, projects, or discussions) when teaching about water? Please give examples.
3. Have you ever use STEM teaching practices in your classroom?
4. Have you ever try integrated teaching? (prompt)
5. What kind of materials or resources (textbooks, charts, videos, etc.) do you use when teaching these topics?
6. Do you talk with students about saving water in their daily life? If yes, how?
7. Have you ever taught about countries in world facing severe water crises and climate change? (prompt)
8. Have you noticed any water conservation practices happening in the school (for example, students not wasting water, posters in school, reminders to save water)?
9. Do students show interest when you teach about floods, irrigation, pollution, or water use? How do they usually respond?
10. What challenges do you face in teaching these topics? (e.g., lack of time, resources, student interest)
11. In your opinion, what else can schools do to make students more aware about saving water and protecting the environment?

Appendix-C

Student Focus Group Discussion (FGD) Tool

A. Students' Knowledge

1. What do you already know about saving water in your home or school?
2. Can you tell me the names of water sources?
3. Can you explain what you know about the water cycle? (like evaporation, condensation, rainfall and runoff)
4. Why do you think rivers are important for people, animals, and the environment?
5. Have you heard about floods, droughts, or water pollution? What do you know about them?
1. Do you know about irrigation in Pakistan? What good things and problems can it bring?

B. Classroom Teaching Practices

1. How does your teacher teach about water, rivers, or the environment? (like stories, pictures, experiments, or activities)
2. Does your teacher give real-life examples from your community or from Pakistan? Can you share one?
3. Do you do activities in class about water and rivers? (like posters, projects, experiments, or group work)
4. Do your teachers talk about how you can save water in your daily life?
5. Which teaching methods or activities do you enjoy the most when learning about water and rivers? Why?
- 6.

C. Students' Suggestions

1. What more would you like your teachers or school to do to make learning about water and the environment more interesting?
2. If you had to give your family or friends one important message about saving water, what would it be?

Appendix-D
CONSENT LETTER

**Government College of Education**
Block 15, F. B. Area, Karachi
DDO code KQ-2149

NO. GCE/ - / 2024-2025 Dated: 27/09/25

The H.M/Principal,
Ibrahim Ali Bhai GGS,
FB-Area BL# 15

Subject: PERMISSION TO CONDUCT ACTION RESEARCH AS PART OF
B.Ed. (Hons.) 4 YEAR (ELEMENTARY) 8TH SEMESTER

Dear Sir/Madam,

The action research is an extended activity of the teaching practice (Practicum) of the current B.Ed. curriculum. Mr./Ms. Maleeha Khan is currently enrolled in this college as a student of B.Ed. (Hons.) 4 year (Elementary) 8th semester 2025. He/She is conducting a research entitled Encouraging water conservation in Schools;
Testing the impact of awareness campaign under the supervision of Prof. Shagufta khalcar

Kindly allow him/her to implement innovative teaching strategies and/or to collect necessary data. During this process, he/she will strictly follow the research ethics. Your cooperation and facilitation in this academic activity will be highly appreciated.

Best Regards


PRINCIPAL

For
Head Mistress
Ibrahim Ali Bhai Elementary
Girls School No.4,
Gulberg Town, Karachi.